

Reducing clients' influence in property valuation: An exploration of a blockchain-based solution

Chibuikem Michael Adilieme^{*}, Rotimi Boluwatife Abidoye, Chyi Lin Lee

School of Built Environment, Faculty of Arts, Design and Architecture, UNSW Sydney, Australia

ARTICLE INFO

Keywords:

Blockchain
Clients influence
PropTech
Property valuation
Property valuer
Nigeria

ABSTRACT

Clients' influence remains a significant issue in the property valuation industry, particularly in emerging countries like Nigeria, where existing solutions have proven ineffective. This study, therefore, considers blockchain technology as a medium for delivering information transparency and explores the application of a blockchain-based tool for reducing clients' influence in property valuation. Specifically, this study is pioneering in proposing, developing, and testing a blockchain-based solution for reducing client influence in property valuation, suggesting a promising avenue for increasing transparency in the process. This study involves several stages. Firstly, *propchain*, a blockchain-based mobile application, was developed and deployed on the Hyperledger Fabric Network, a permissioned blockchain network. The efficacy of the proposed solution was qualitatively tested by a single-round focus group held with valuers, bankers, and a loan customer who used the application to simulate a property valuation exercise for a mortgage. The findings suggest that blockchain technology could be a viable tool for reducing client influence in property valuations. However, specific challenges in the Nigerian context, such as data access, information verification, and stakeholder trust, present significant obstacles. These issues highlight the need for robust institutional support systems to enhance the efficacy of blockchain solutions in this domain. This study supports Levy and Schuck's (2005) assertion that increased information transparency can reduce client influence. Nevertheless, our research indicates that the successful implementation of blockchain in property valuation will require overcoming systemic barriers unique to the Nigerian environment. The implication of key findings is also discussed.

1. Introduction

Research on the clients' influence problem in property valuation has existed since the late twentieth century (Adilieme et al., 2023; Kinnard et al., 1997; Smolen & Hambleton, 1997). The problem doesn't appear prominent in recent literature on developed property markets (Cheloti & Mooya, 2021; Crosby et al., 2018). This may be due to regulatory frameworks and mechanisms such as rigorous standards, market transparency, professional oversight, institutional checks (Lee & Lee, 2014; Zhang et al., 2019) designed to enhance valuer objectivity. However, it still appears in literature to be a challenge in developing and less mature property markets such as Nigeria and Kenya, to name a few (Adilieme et al., 2024a; Cheloti & Mooya, 2021, 2023; Nwuba et al., 2015a). Whilst no study has been put forward for the sole purpose of developing and/or testing solutions to address the issue of clients' influence in property valuation, Adilieme et al. (2023) highlights several studies that

have made suggestions to address the issue, with focus weighing towards property valuers.

Clients' influence refers to actions by clients that interfere with the valuation process and lead to a biased property valuation outcome (Achu et al., 2015; Nwuba, 2015b). The suggestions to address the clients' influence challenge include review and enforcement of regulations, adherence to professional ethics and standards, increasing valuer income and introduction of a peer review and independent audit system (Amidu & Aluko, 2007a; Kinnard et al., 1997; Nwuba et al., 2015a, 2015b; Smolen & Hambleton, 1997). Additionally, Amidu and Aluko (2007b) argue in favour of formalising the valuation process to clarify the expectations and deliverables, which reduces negative forms of client influence. Other suggested solutions include targeting the valuation environment through improved information transparency (Chen & Yu, 2009; Levy & Schuck, 2005). Furthermore, Freybote et al. (2014) provide evidence from the USA that introducing information

^{*} Corresponding author.

E-mail addresses: c.adilieme@unsw.edu.au, adiliememichael@gmail.com (C.M. Adilieme), r.abidoye@unsw.edu.au (R.B. Abidoye), chyilin.lee@unsw.edu.au (C.L. Lee).

<https://doi.org/10.1016/j.habitatint.2024.103217>

Received 16 June 2024; Received in revised form 31 October 2024; Accepted 3 November 2024

Available online 7 November 2024

0197-3975/© 2024 The Author(s). Published by Elsevier Ltd. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

intermediaries between valuers and lenders reduced clients' influence, thereby crediting the suggestion of information transparency. Information transparency is a logical suggestion due to the importance of quality information to the property valuation process, and some clients leverage this by supplying unreliable information (Bellman, 2018; Levy & Schuck, 2005). Hence, valuers are expected to verify the information used in the process (Klamer et al., 2019).

Despite the numerous suggestions made, recent studies in developing and less transparent property markets such as Nigeria and Kenya reveal that clients' influence is still a significant problem befuddling the property valuation industry (see Adilieme et al., 2024a; Cheloti & Mooya, 2023). This does not imply that those property markets have not implemented the suggested solutions, such as using regulations and peer review systems. However, compared to developed property markets, those markets are not well-developed and structured to effectively manage the valuation problems (Bellman, 2018; Crosby et al., 2018). Hence, valuers in those sorts of markets with limited institutional capacity and challenges usually employ different creative means to surmount practice challenges (Ansenberg et al., 2024). In addition to the lack of institutional capacity to deliver solutions, information transparency has not been tested in these markets.

The property sector has seen a growing application of property technology (PropTech), which encompasses innovations such as artificial intelligence (AI), machine learning (ML), and blockchain (Taecharungroj, 2021; Iban, 2022; Jafary et al., 2024). These technologies transform and automate property valuation and management processes and address broad housing market issues like gentrification, value drivers (Taecharungroj, 2021; Jafary et al., 2024; Soltani & Lee, 2024; Thackway et al., 2023a, 2023b). Despite the widespread use of blockchain technology in various sectors such as finance, construction, logistics, and healthcare (see, Xu & Zou, 2021; Hamledari & Fischer, 2021; Orji et al., 2020; Leite et al., 2021) to introduce transparency, its application in the property valuation sector is still nascent and remains largely underexplored (Adilieme et al., 2024b). Consequently, this study proposes and tests using a blockchain-based application to reduce clients' influence in property valuation, with Nigeria as a case study. Blockchain is a decentralised database existing across several locations and among multiple participants (Baker et al., 2021; Liu et al., 2020). It can record transactions and share across a network of users (Mashatan et al., 2021). A blockchain consists of four primary characteristics: a distributed peer-to-peer (p2p) network, consensus or trust-based digital transactions, tamper resistance and a stored ledger for record-keeping (Baker et al., 2021; Bhatia et al., 2020). Hence, these features make it an intermediary that delivers information transparency, as proposed by Levy and Schuck (2005).

Studies such as Veuger (2018), Wouda and Opdenakker (2019), Pongnumkul et al. (2020, pp. 100–104), Ameyaw and de Vries (2021) have identified prospects for the application of blockchain in different property transactions (land registration, property acquisition and investment) due to its ability to increase information transparency, enforce smart contracts, prevent frauds and promote efficient data management and have developed theoretical and conceptual frameworks to advance their claim. However, despite its acknowledged prospects, little to no actual use of the technology has been observed (Adilieme et al., 2024b). Hence, to support the study's aim of exploring using blockchain technology to reduce clients' influence in property valuation, this study sets out to first develop a blockchain-based mobile solution, use it to simulate a valuation process, and assess whether there was transparency and whether it effectively reduced or eliminated clients' influence.

This study contributes to the literature twofold. Firstly, it is the first study to propose and test a blockchain technology-based solution to the clients' influence problem. Secondly, this study assesses if blockchain can introduce information transparency and potentially modify the valuation process in Nigeria. Nigeria, as a case study, is based on two premises. Firstly, the persistent issue of clients' influence, as noted

earlier. Secondly, based on its position as a leading property market in Africa, with a property market value of US\$2.1 trillion in 2023 (Statista, 2024), poor operations in its property industry could reverberate across the region.

2. Literature review

2.1. Addressing the clients' influence challenge – suggested strategies

Property valuation is required by individuals, investors and organisations within the global economy for various purposes because of limited access to real-time information on the property market (Crosby et al., 2019; Klamer et al., 2021). Consequently, property valuation outcomes guide decisions and influence outcomes such as share prices of REITs, wages and bonuses, and access to credit (Crosby & Devaney, 2019; McAllister, 2019). As a result, some stakeholders attempt to impose their will on the property valuation process to get a favourable outcome. In addition, in some climes, clients serve as sources of information; in other climes, they are provided with interim reports to reflect, whilst, in some markets, they have huge power in the valuer selection process, all of which allows them to try to impose their will (Klamer et al., 2018; Levy & Schuck, 2005; McAllister, 2019). Ultimately, these motives, actions and means by clients on the property valuation process have led to some negative consequences globally, both within the financial market and property market (Augustyniak et al., 2018; Crosby et al., 2019; Gilbertson & Preston, 2005; Newell et al., 2010; Rudolph, 1998).

Graaskamp (1988), as cited in Kinnard et al. (1997), advocated for federal regulation in the property valuation industry in the USA to address valuer behaviour by eliminating valuers pliant to clients. Additionally, Smolen and Hambleton (1997) argued for the use of practice licences for valuers, retraining of key finance sector players, audit of mortgage bankers to control the mechanism for selecting valuers, peer review of valuers, bonding certified valuers, ensuring complete independence for in-house valuers, and lowering the *de minimis* for property classes. At the same time, Kinnard et al. (1997), also in the USA, found that regulation through certification and licensing was ineffective in reducing clients' influence and increasing valuer independence. Accordingly, Kinnard et al. (1997) suggested using independent auditors and diversification of the client base as a possible solution. In contrast to the findings of Kinnard et al. (1997) on the limited effectiveness of regulations through certification and licensing, Freybote et al. (2014) found that the Dodd-Frank Act reduced clients' influence as it introduced third-party intermediaries such as appraisal management companies (AMCs) between valuers and lenders, to coordinate the appraisal process thereby eliminating direct contact and feedback (Freybote et al., 2014; Sanderford et al., 2017). The findings by Freybote et al. (2014) indicate how legal and regulatory tools can support solutions to reduce clients' influence. In the study of Freybote et al. (2014), the law itself did not reduce the issue but created intermediaries that maintained information transparency between stakeholders.

Despite the acknowledged positive performance of the Dodd-Frank Act in reducing clients' influence in the USA, criticism about the increased cost, time and impact on a mortgage loan process has emerged. Holt et al. (2015) argue that whilst the Act served the purpose of increasing transparency and trust in the appraisal process for mortgages, an unintended consequence is the cost accrued to lenders, valuers and clients due to the operations of the AMCs that sees valuers travelling to far areas to cover valuation thereby leading to increased turnaround time and cost to clients and lenders. Ding and Nakamura (2016) suggest that while the efforts of the Home Valuation Code of Conduct (HVCC) now contained in the Dodd-Frank Act reduced artificially inflated valuation figures, it has produced some externalities such as a conservative approach to valuation by valuers, resulting in low appraisal figures which have made the mortgage process more difficult.

In Nigeria, Amidu and Aluko (2007a) proposed creating a

monitoring system by valuation organisations to ensure valuers comply with the code of conduct. Likewise, [Amidu and Aluko \(2007b\)](#) suggested formalising the valuation process from instruction to reporting to protect against clients' coercive powers. Also, in Nigeria, [Nwuba et al. \(2015a\)](#) and [Nwuba et al. \(2015b\)](#) advocated for the ethics and integrity of the valuer backed by practice standards and regulations. Like the suggestions made for valuers in Nigeria, [Kucharska-Stasiak et al. \(2018\)](#) suggested reviewing and enforcing a professional code of conduct for valuers in Poland. Whilst in Taiwan, [Liao et al. \(2018\)](#) recommended an independent audit system overseen by the valuers' association. [Lee et al. \(2020\)](#), also using data from Taiwan, suggested increased income and public recognition of the fairness of valuers as a possible way to improve their objectivity. These suggestions, especially from Nigeria, have not resolved the issue, as its persistence is highlighted in the earlier section of this study.

From the literature review, most suggestions lean towards measures involving regulations and legal support. However, studies exist that consider other forms of solutions. [Levy and Schuck \(2005\)](#), focusing on New Zealand, highlighted the importance of clients as information suppliers with expertise and experience; hence, they suggested that a solution could be to separate and/or limit the contact between valuers and clients. [Levy and Schuck \(2005\)](#) further argued that greater transparency and higher levels of disclosure could reduce clients' influence. The suggestion of information transparency by [Levy and Schuck \(2005\)](#) could be argued to have been validated by the findings of [Freybote et al. \(2014\)](#) on the successful use of AMCs as information intermediaries to promote information transparency.

Though not expressly suggesting a solution, [Chen and Yu \(2009\)](#) comparatively analysed data from Taiwan and Singapore and posited that information transparency reduced the incidence of clients' influence. The findings by [Chen and Yu \(2009\)](#), like [Freybote et al. \(2014\)](#), seemed to confirm [Levy and Schuck's \(2005\)](#) position on the place of information transparency in reducing clients' influence.

From the foregoing, the suggestions have reinforced valuer integrity and ethics, enforcement of standards and a peer review/audit system to act as a check for valuers. Furthermore, these systems could be used to promote information transparency, as highlighted in the USA. However, these solutions may not hold the same efficacy for all property markets; as [Bellman \(2018\)](#) and [Kucharska-Stasiak et al. \(2018\)](#) argue, some property markets may be unable to monitor and regulate valuers effectively.

Accordingly, increasing transparency for the stakeholders may be one option left for those property markets with limited governance and monitoring capacity. Furthermore, given the criticism raised about the effect of intermediaries on the valuation process, to curtail such externalities, the option of transparency is explored in this paper by utilising blockchain, a technological tool, to test as a solution to reduce clients' influence in Nigeria.

Some examples abound in the proposal of blockchain technology to address issues in the built environment and property. [Pradeep et al. \(2021\)](#) developed a tool for information transparency in the construction industry using blockchain. [Werbach \(2018\)](#) reports on a blockchain property registry for Cook County, Illinois; however, [Adilieme, Abidoye, and Lee \(2024c\)](#) critiques the study as lacking in detail about the property registry. Similarly, [Konashevych \(2021\)](#), reviewing a blockchain property registry in Afghanistan, critiques studies that tout the use of blockchain for addressing property challenges as not revealing how such solutions will be designed or implemented.

From the foregoing, whilst there is no retinue of empirical studies on using blockchain technology for property valuation, there exists a proof of concept from Hong Kong that utilised blockchain technology as a tool to facilitate a mortgage application process that included property valuation components (see, [Hong Kong Monetary Authority, 2016](#)).

2.2. Blockchain: A governance tool for transparency and trust

As a type of digital technology, blockchain has been touted to be a governance tool that, amongst other things, can reduce the monopolistic powers of parties in a transaction through increased transparency ([Hooper & Holtbrügge, 2020](#)). [Werbach \(2018\)](#) posits that "the basic function of blockchain is to reliably share information among parties who may not trust one another. In other words, everyone can have his or her copy of a ledger and trust that all those copies remain the same, even without a central administrator or master version" (p.7). [Konashevych \(2021\)](#) argues that the potential for blockchain technology as a governance tool has been explored by both academia and governments since around the late 2010s.

[Mashatan et al. \(2021\)](#) argue that blockchain is an alternative tool for recording transactions. Transactions on the blockchain are visible to all participants and incentivise honest behaviour, as dishonest behaviour will be evident to all participants, thereby ensuring no participant can influence a blockchain-based system ([Mashatan et al., 2021](#)). Accordingly, Blockchain is deemed a social technology designed to govern the behaviour of groups of people through social and financial incentives ([Graglia & Mellon, 2018](#)). [Werbach \(2018\)](#) argues that blockchain technology could function as a governance tool, both in a supplementary and complementary form, to support legal tools. In areas where legal enforcement is weak, blockchain could take its place as an alternative tool ([Werbach, 2018](#)).

The potential for governance and transparency through blockchain has been trialled in property administration in some climes. [Werbach \(2018\)](#) reports a pilot test by Cook County, Illinois, on the successful development of a blockchain-based property registry. [Shang and Price \(2019\)](#) report on the pilot test of a blockchain-based land titling system in the Republic of Georgia, which introduced a semblance of trust in the title system, though with some public scepticism. [Graglia and Mellon \(2018\)](#) report on the development and testing of blockchain for property administration in Brazil, Georgia, Dubai, and Sweden. The pilot tests provided results that revealed the potential for blockchain to make property transactions and records transparent. An observation from these studies is that there is no widespread evidence of its adoption as a governance tool beyond property administration. Furthermore, cautious projections about the efficacy of blockchain for governance in property transactions and administration need to be made, as [Konashevych \(2021\)](#) argues that a pilot study on blockchain-based property systems in Afghanistan did not show success in replacing existing traditional systems.

[Hong Kong Monetary Authority \(2016\)](#) reports a blockchain technology proof of concept for the mortgage process, which resulted from a collaboration between the Hong Kong Applied Science and Technology Research Institute and some financial institutions. This process included a property valuation; the brief and reports were communicated via the blockchain. However, much attention was not given to the property valuation component and whether blockchain was explicitly incorporated to address a property valuation concern.

[Davis et al. \(2022\)](#) acknowledge the potential for blockchain to curb corruption through transparency but argue that the outcome of the technology's application is not a predictable certainty since it will depend on the intent and mode of application. Hence, blockchain-based solutions need to consider the type of blockchain and the participants ([Davis et al., 2022](#)). Accordingly, the remaining literature review addresses the selection of a blockchain platform to create a blockchain-based solution.

2.3. Selecting a blockchain platform

Many blockchain platforms with different capabilities and features have been introduced by developers ([Nanayakkara et al., 2021](#)). They include Corda, Ethereum, and Hyperledger Fabric, among others ([Yang et al., 2020](#)). [Ahmad et al. \(2021\)](#) compared seven blockchain platforms

for utilisation in developing solutions applicable to logistics. The platforms include Hyperledger Fabric, Hyperledger Iroha, Hyperledger Sawtooth, Ethereum, Quorum, Hyperledger Besu and OpenChain (Ahmad et al., 2021). Nanayakkara et al. (2021) identified and compared twenty-four blockchain platforms for use in developing enterprise solutions.

Nanayakkara et al. (2021) argue that due to the plethora of blockchain platforms, there is difficulty in selecting a suitable blockchain platform. Hence, Nanayakkara et al. (2021) propose a four-stage methodology to identify, select, evaluate, and validate the blockchain network platform. Perera et al. (2021) suggest that developing a blockchain system follows a process of first analysing the problem and then selecting the blockchain platform. In addition, Nanayakkara et al. (2021) and Perera et al. (2021) recommends a Simple Multi Attribute Rating Technique (SMART), a multi-criteria decision-making technique, to assess the different blockchain networks through an analysis of their system architecture, libraries, tools, applications and capability. This SMART technique can be used to decide on the blockchain platform for developing a solution. Accordingly, twenty-six blockchain platforms have been identified and presented in Table 1. Table 1 shows blockchain platforms, network types, and the programming languages they support for building applications and other solutions.

Among the different blockchain platforms, Perera et al. (2021) advocate for permissioned blockchain for property transactions as it identifies participants while ensuring the security of sensitive user data. Furthermore, when identified, participants will be held accountable for transactions instead of systems where participants operate anonymously. In addition, Perera et al. (2021) highlight that the Hyperledger Fabric network seems a popular choice for use within the built environment.

The perceived appeal for the Hyperledger Fabric network to develop solutions for the built environment seems credible. Wang et al. (2020) built a solution for supply chain traceability and information sharing. Wu et al. (2022) developed a solution to ensure safety inspection of tower cranes. Hijazi et al. (2023) developed a proof of concept for integrating BIM and blockchain into a single source of truth for construction supply chain data delivery.

Another blockchain platform that has also seen considerable adoption for testing solutions is the Ethereum platform. Yang et al. (2020) utilised the Ethereum and Hyperledger Fabric to develop a solution for the construction industry. Hamledari and Fischer (2021) built a construction payment solution using the Ethereum network. However, Hijazi (2023), citing Nawari (2020), argued that the Hyperledger Fabric blockchain network is better suited for application to the built environment due to its design and implementation flexibility and confidentiality. Furthermore, using their SMART analysis, Nanayakkara et al. (2021) identified the Hyperledger Fabric as the optimum platform for building enterprise applications due to its higher security, privacy, and participant confidentiality. Furthermore, Perera (2021) selects the Hyperledger Fabric blockchain network as suitable for developing solutions for property transactions after considering the learning material and support, cost, ease of use and language, performance, permissioned feature, reputation, security, and integration with applications. Consequently, based on its popularity and permissioned nature, which ensures the identification of participants and data security, this study selected the Hyperledger Fabric blockchain network to develop a property valuation application.

The following segment discusses the approach to developing the blockchain-based solution, testing it, and validating its use in achieving this study's aim of reducing clients' influence in property valuation.

3. Material and methods

This study adopted a combination of design science and qualitative methods to achieve its aim. The design science approach was used to develop a blockchain-based application, and qualitative methods were

Table 1

Comparison of different Blockchain platforms based on their features.

Blockchain platforms	Network type	Supported languages	Cost Efficiency	Source(s)
BigChainDB	Permissionless	JavaScript, Python	Low	Nanayakkara et al. (2021)
Chain Core	Permissioned	Java, Ruby	High	Nanayakkara et al. (2021)
Corda R3	Permissioned	Kotlin, Java	Low	Nanayakkara et al. (2021)
Cosmos	Permissioned, Permissionless	Golang	High	Cosmos (2023); Ledger Insights (2019)
Credits	Permissionless	Java	Low	Nanayakkara et al. (2021)
Elements	Permissioned	Python, Go, C#, Java, Ruby, Perl	High	Nanayakkara et al. (2021)
EOSIO	Permissionless, private	C++	Low	Nanayakkara et al. (2021)
Eris	Permissioned	Solidity	High	Nanayakkara et al. (2021)
Ethereum	Permissionless	Solidity, Vyper	Low	Ahmad et al. (2021); Nanayakkara et al. (2021)
Hydrachain	Permissioned	Python	High	Nanayakkara et al. (2021)
Hyperledger Besu	Consortium	Solidity	High	Ahmad et al. (2021)
Hyperledger Fabric	Permissioned	Golang, Java	High	Ahmad et al. (2021); Nanayakkara et al. (2021)
Hyperledger Iroha	Permissioned	Java, C++	High	Ahmad et al. (2021); Nanayakkara et al. (2021)
Hyperledger Sawtooth	Permissioned	Python, Rust, JavaScript, Go, C++, Solidity	High	Ahmad et al. (2021); Nanayakkara et al. (2021)
IBM Blockchain	Permissioned	JavaScript, Java, Go, Python, Node.js	Low	Nanayakkara et al. (2021)
IOTA	Permissionless	JavaScript, Kava, C, Go, Python	Low	Nanayakkara et al. (2021)
Lisk	Permissioned	JavaScript	High	Nanayakkara et al. (2021)
Multichain	Permissioned	Python, C#, Java	Low	Nanayakkara et al. (2021)
NEM	Permissioned, Permissionless	Java, C++	Low	Nanayakkara et al. (2021)
NEO	Permissionless	C#, Python, Java, Go	Low	Nanayakkara et al. (2021)
OpenChain	Permissioned, Permissionless	Java, Clojure, Groovy, Kotlin, Scala, JavaScript, C++	High	Ahmad et al. (2021); Nanayakkara et al. (2021)
Quorum	Permissioned, Permissionless	Solidity	High	Ahmad et al. (2021); Nanayakkara et al. (2021)
Stellar	Permissionless	JavaScript, Java, Go, Ruby, Python, C#	High	Nanayakkara et al. (2021)
Stratis	Permissioned	C#	Low	Nanayakkara et al. (2021)
Symbiont Assembly	Permissioned	Python, SympL	Low	Nanayakkara et al. (2021)
Waves	Permissioned, Permissionless	Ride	Low	Nanayakkara et al. (2021)

Source: Authors compilation

used to validate its use in reducing clients' influence in property valuation.

3.1. Design science method

Van Aken and Romme (2012) describe design science as a family of research approaches driven by field or real-life problems and adopts a solution-based orientation to address those problems. Van Aken and Romme (2012) explain that for design science, the researchers go beyond describing field problems and their causes to developing alternative solution concepts for the problems. A design science approach typically involves problem identification, solution design/development, demonstration and evaluation, and communication of the results (Peffers et al., 2007). This approach is typical in information systems, with the development of software and applications to address research or practical problems (Peffers et al., 2007). Furthermore, the design science approach has been used in developing practical models and solutions in spatial planning (Yilmaz et al., 2023), real estate (Grant & Cherif, 2016), life sciences (Maphosa, 2022; Wendimu & Biredagn, 2023). Wu et al. (2022) applied the design science approach to develop a blockchain-based prototype for on-site safety inspection of tower cranes. In addition, studies such as Wyatt (1996, 1997) developed GIS-based property information systems for property valuation, giving credence to developing and using technology artefacts to address property valuation challenges. Accordingly, a blockchain system was developed on the Hyperledger Fabric network, and an Android mobile application, *propchain*, was developed to interact with the blockchain network. The application was preferred because the participants mainly relied on their mobile phones for internet connectivity; this is a trend typical in Nigeria as mobile devices are more used than desktop devices for internet access (Sasu, 2023).

3.2. Qualitative method - focus group

To test the developed solution, the *propchain* was distributed to five participants listed in Table 2: two valuers (PV1 & PV2), two bankers (B1 & B2) and one client/user - customer seeking credit from the banks (C1), whilst the research team were the administrator(s) of the application. The focus group lasted about 2 h, and all the participants had an opportunity to reflect on the *propchain* and discuss its use for reducing clients' influence.

Seventeen participants (twelve valuers and five bankers) were considered. This study is a component of a research project on clients' influence. Hence, the participants in this stage were drawn from participants in the early parts of the research, selected through a snowball approach and interviewed on different perspectives on the clients' influence problem in property valuation in Nigeria and the valuation industry in Nigeria. After that, two valuers and two bankers were selected from the pool of participants based on their experience and familiarity with using mobile technology systems to conduct property valuation. An independent participant was included as the credit-seeking customer. These participants simulated a property valuation process for a typical residential property for mortgage purposes, wherein a credit-seeking customer requires a loan and approaches a bank. The bank demands a

property valuation report from a property valuer through its official.

The application was validated through the qualitative approach of a single-round focus group session with the participants via Zoom. Smithson (2000) argues that focus groups are interactive discussions within a specific, controlled setting. Parker and Titter (2006) explain that focus groups emphasise the inter-relational dynamics of participants and the ideas their facilitated discussions generate. Hence, the focus group session allowed participants to collectively reflect and bring their unique perspectives to assess the blockchain application for the study's goal. This confers an advantage of providing the study with a collective group opinion. It also made it possible to explore how the norms of the different property valuation actors played out within a typical setting in line with the findings of Guest et al. (2017) that certain disclosures and themes most likely emerge in focus groups.

The choice of qualitative assessment was conceived on two premises. Firstly, blockchain is a relatively unexplored tool in property valuation research and practice globally and in Nigeria. Hence, consideration was given to ease of access for willing participants. Secondly, from the extant client influence in property valuation literature, there are no quantitative measures for assessing transparency, as suggested by Levy and Schuck (2005). Therefore, to validate the application, the discussions among the participants revolved around the following elements. Firstly, the usefulness of a blockchain-based application in the property valuation process within the context of the clients' influence problem. Secondly, the ease of use of such a solution. Thirdly, if the *propchain* changed the approach to a valuation assignment. Fourthly, the level of confidence in the ability of the *propchain* to reduce clients' influence. Finally, barriers or challenges perceived within the context of the clients' influence problem.

3.3. Developing the blockchain network and mobile application

Following the literature review on selecting a blockchain platform that fits the solution type and participants, the blockchain network was designed based on the workflow in Fig. 1. This workflow mimicked the typical valuation process for mortgage purposes as informed by valuers and bankers in Nigeria. The operation is such that a loan-seeking customer (secondary client) approaches a bank for credit. The bank (primary client) assigns a valuation request to a valuer through its designated official. The valuer conducts the valuation and submits the report. Accordingly, two clients with competing interests exist in the typical property valuation transaction involving financial institutions (see, Adilieme et al., 2024a). However, blockchain technology offers a solution to mitigate these conflicting interests. By providing a transparent and immutable record of all communications and data exchanges between the valuer, property owner, and lender, blockchain ensures that all parties have access to the same information. This transparency reduces the potential for bias or manipulation in the valuation process, helping to align the interests of the property owner and the lender and ensuring a fair and accurate outcome for both parties.

Consequently, in the context of the property valuation process, blockchain technology will sit as a communication and storage centre that facilitates the communication between all parties and supports the dissemination of documents and other information (see, Hong Kong Monetary Authority, 2016). Furthermore, the *propchain* is coded to prevent revising reports. Hence, once a report is submitted, it can only be resubmitted upon validation by all parties.

Levy and Schuck (2005) and Chen and Yu (2009) propose information transparency as a tool to reduce clients' influence since it reduces the probability of clients leveraging information and procedural powers. The information transparency proposition is validated in Freybote et al. (2014) on the role played by AMCs, which reduced clients' influence. Hence, given the highlighted institutional gap in Nigeria, the expectation is for blockchain technology, which has been touted as decentralised and transparent, to be a trusted intermediary for facilitating real-time communication and tracking, guaranteeing information

Table 2
Focus group participants.

Participant	Academic qualification	Role	Years of experience
PV1	Master's Degree	Associate partner (valuation practice)	18
PV2	Postgraduate Diploma	Principal Partner (valuation practice)	25
B1	MBA	Assistant Manager	19
B2	MBA	Branch Manager	15
C1	Master's degree	Property consultant	4

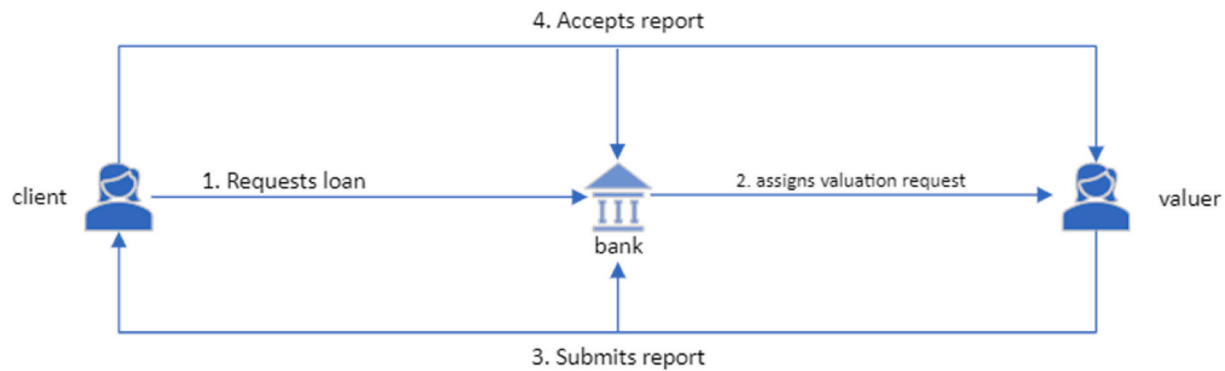


Fig. 1. Illustration of the workflow for the proposed blockchain application.

transparency.

The blockchain system was developed on the Hyperledger Fabric client Software Development Kit (SDK). It consisted of one Certificate Authority (CA) for administration, one ordering service (orderer), two organisations, and two peer nodes (peers) to support the two bankers, two valuers, one credit-seeking customer, and the backend administrator. These system components are shown in Fig. 2, a screenshot of the docker container.

Upon development, the blockchain was deployed on an AWS EC2 Ubuntu machine, as shown in Fig. 3 and connected to an AWS S3 bucket to store uploaded files and documents. Smart contracts (chaincode in Hyperledger Network) were written using Java and installed to guide and automate actions or events on the propchain.

Fig. 4 shows a snapshot of a smart contract that requests selected property details and documents to be submitted via the propchain. Figs. 5 and 6 show snapshots of smart contracts controlling enrolment and permissible operations on the propchain. The entire smart contract and other attendant codes and files for the blockchain system and mobile application are accessible online.¹

3.4. Using and testing the blockchain-based application

Fig. 7 shows a screenshot of the propchain in use. From the screenshot, the first image shows the enrolment page where the user enrolls with a username and password and is authenticated by selecting a role between administrator (admin), banker (bank), valuer, and client (credit-seeking customer). The enrolment ensures the identification and security of the parties in line with the permissioned nature of the Hyperledger Fabric network. The second image and third show the participants' views when a valuation request has been made and concluded, with all parties able to download submitted documents without needing any off-chain communication.

Fig. 8 shows the screenshot taken via Windows PowerShell to display a completed transaction that formed a validated block in the blockchain with the date and time stamp in blue. This indicates that the transaction is permanently on record and cannot be changed or disputed as it captures exactly when it was performed on the application.

Note herein that from Fig. 8, what stands out is that the transactions on the blockchain network are recorded as hashed codes. This implies that while from the backend, there is full knowledge of a completed transaction, details of the content of that transaction are only captured in the front end on the application of the participants and any storage systems (if applicable).

3.5. Validating the propchain

Upon completion of the simulation of the valuation process using the

propchain, the focus group session was held. Transcripts of the recorded session were analysed using the NVivo 12 software to identify common themes and insights on the effectiveness of blockchain in reducing clients' influence in property valuation. This approach of identifying, analysing, and reporting data patterns is called thematic analysis (Braun & Clarke, 2006). The researchers also reviewed the transcripts to ensure all themes were identified. Accordingly, the themes that emerged include blockchain technology as a useful tool to reduce clients' influence, the blockchain-based application as an easy tool to use, blockchain as a potentially modifying approach to a valuation assignment, barriers to the use of blockchain for valuation, and ways to promoting the use of blockchain for property valuation.

4. Results

Five themes emerged from the data analysis. They include blockchain as a useful tool for reducing clients' influence, ease of use, modifying the approach to valuation, issues limiting effectiveness, and adoption mechanisms.

4.1. Theme 1: Blockchain technology, a useful tool for reducing clients' influence in property valuation

In assessing the use of Propchain in conducting valuation, the participants provided perspectives and reflections that showed positive sentiments for integrating blockchain technology into the property valuation process as an intermediary for information transparency and to achieve the study's goal of reducing clients' influence.

A key statement was, "From a Nigerian perspective, the whole process of using the blockchain was straightforward, and I perceived that it would eliminate client influence, which is very good." (B1). Another important statement affirmed and reflected was, "The blockchain application for valuation report is useful and could be instrumental to an improved regulatory system if adopted. This can help improve transparency and limit client/mortgagor influence." (PV2).

The participants' views aligned with the position of Pradeep et al. (2021) on blockchain providing supplementary support to existing systems. Nevertheless, the group raised and affirmed concerns about the overall effectiveness of blockchain in reducing clients' influence. Some of these include:

"The propchain may not necessarily be completely effective because I still see an avenue for human interaction, and the fact that there will be contact with the property owner during the inspection is a potential problem. Notwithstanding, it gives a sense of monitoring, which will reduce the likelihood of interference. Though I think the key to limiting interference is the adoption of disciplinary measures strictly." (PV2).

The issue pointed to the fact that clients could influence the valuers during the inspection stage by uploading misleading information to the application, especially in Nigeria, where clients are a huge source of

¹ Propvalchain files.

```

https://ubuntu.com/aws/pro

Expanded Security Maintenance for Applications is not enabled.

98 updates can be applied immediately.
1 of these updates is a standard security update.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

1 updates could not be installed automatically. For more details,
see /var/log/unattended-upgrades/unattended-upgrades.log

*** System restart required ***
Last login: Thu Nov  9 21:59:32 2023 from 149.171.161.111
ubuntu@ip-172-31-29-215:~$ docker ps
CONTAINER ID   IMAGE                                COMMAND
eeab175ce672   dev-sample-corp-two.propvalchain.com-production-070be83558fb2d7ef743f040e5520c7bd2f35a60c632e18be2ffaa6809fb7a5c-00aa17f47ad06cab952a501445c8a7110cb10c0d4d43e7ebef99dc210bd30  "/root/chaincod
e-jav.-" 6 months ago Up 6 months
35f6cfaba391   dev-sample-corp-one.propvalchain.com-production-070be83558fb2d7ef743f040e5520c7bd2f35a60c632e18be2ffaa6809fb7a5c-3c3e9aa530396f2869c79805aa4cb5d04b63c88658b4cf7ff440ad099a39cf1  "/root/chaincod
e-jav.-" 6 months ago Up 6 months
65816f596c87   dev-sample-corp-two.propvalchain.com-gama-84d62b81286099aa8f219b58d931737dd131071110b7ce4910b27625dc68099-4468d44c2afdc1dd8f56664aa68ea824e35898563be1380f878f8ea79d2a19d  "/root/chaincod
e-jav.-" 6 months ago Up 6 months
ad69c278f93c   dev-sample-corp-one.propvalchain.com-gama-84d62b81286099aa8f219b58d931737dd131071110b7ce4910b27625dc68099-2aa766e5e56d7e51f3229ee79a46c4aed50d73976cc162315d6dc7c44fa953a  "/root/chaincod
e-jav.-" 6 months ago Up 6 months
e1a367dfff45   dev-sample-corp-two.propvalchain.com-beta-df873372c8e134c69833680f506e5b0aff704080373aa36269680a89cf4f141-20345f912d6e54d0bcf85ab8d83345166c14e8e885f38872218c9385d5f229ae  "/root/chaincod
e-jav.-" 6 months ago Up 6 months
fa2110b3e3d4   dev-sample-corp-one.propvalchain.com-beta-df873372c8e134c69833680f506e5b0aff704080373aa36269680a89cf4f141-20345f912d6e54d0bcf85ab8d83345166c14e8e885f38872218c9385d5f229ae  "/root/chaincod
e-jav.-" 6 months ago Up 6 months
38f6d1737f35   dev-sample-corp-two.propvalchain.com-test123-1199ccc0494d97234d8a51a92488aeb54965c4af7561956b80b6994ddbe6469a-2adedc36741d4928d201c5a81595b3666e9b23be80b5569e861147f01d762ca0  "/root/chaincod
e-jav.-" 6 months ago Up 6 months
c57e425a5f42   dev-sample-corp-one.propvalchain.com-test123-1199ccc0494d97234d8a51a92488aeb54965c4af7561956b80b6994ddbe6469a-87e99fba71ea23c29f83b6461145bb19ae4f514e760cde62a7a2ec382def938  "/root/chaincod
e-jav.-" 6 months ago Up 6 months
7e31cf65337   dev-sample-corp-two.propvalchain.com-test-8334fa9819fb920a8ee300b17944e6032732ac34249954d9a2f889c11970ad6-165474ff7e72889a6d4fc5c2888b44a45c20c1b7c4234d8bc845639585a86  "/root/chaincod
e-jav.-" 6 months ago Up 6 months
dacf399b6c0   dev-sample-corp-two.propvalchain.com-preprd-6c55d8fc97cee567ae1506877feb5cd8da0e0a54fbc8433c884c2e9c8bf3d7ce-4f85a211f2320dc2a28d65f3181f08792e5e76e5396e9c34ee89a5def61b197  "/root/chaincod
e-jav.-" 6 months ago Up 6 months
666d2ed4f17   dev-sample-corp-one.propvalchain.com-preprd-6c55d8fc97cee567ae1506877feb5cd8da0e0a54fbc8433c884c2e9c8bf3d7ce-4f85a211f2320dc2a28d65f3181f08792e5e76e5396e9c34ee89a5def61b197  "/root/chaincod
e-jav.-" 6 months ago Up 6 months
cccc22195a8c   hyperledger/fabric-peer:2.2.0      sample-corp-two.propvalchain.com
t*              7 months ago Up 6 months 7051/tcp, 0.0.0.0:10051->10051/tcp, :::10051->10051/tcp
724c373aa91b   hyperledger/fabric-peer:2.2.0      sample-corp-one.propvalchain.com
t*              7 months ago Up 6 months 7051/tcp, 0.0.0.0:9051->9051/tcp, :::9051->9051/tcp
149cb4abb010   hyperledger/fabric-orderer:2.2.0   sample-orderer.propvalchain.com
7 months ago Up 6 months 0.0.0.0:7050->7050/tcp, :::7050->7050/tcp
0e2691a09e71   hyperledger/fabric-ca:1.4.7        ca.propvalchain.com
ca-se-         7 months ago Up 6 months 0.0.0.0:7054->7054/tcp, :::7054->7054/tcp
ubuntu@ip-172-31-29-215:~$

```

Fig. 2. Screenshot of the docker container.

Instance summary for i-0677c0067f52844c8 (propvalchain)

Updated less than a minute ago

Instance ID: i-0677c0067f52844c8 (propvalchain)

Public IPv4 address: 3.22.247.157 [open address]

Private IPv4 address: 172.31.29.215

Instance state: Running

Private IP DNS name (IPv4 only): ip-172-31-29-215-us-east-2.compute.internal

Instance type: t3.large

VPC ID: vpc-01144b0e4a404081

Subnet ID: subnet-098f9b9d5c56371

Platform: Ubuntu (Inferred)

Platform details: Linux/UNIX

Stop protection: Disabled

Instance auto-recovery: Default

AMI Launch index: 0

AMI ID: ami-00eedc4036573771

AMI name: ubuntu/images/hvm-ssd/ubuntu-jammy-22.04-amd64-server-20230208

Launch time: Fri Apr 28 2023 01:52:04 GMT+1000 (Australian Eastern Standard Time) [6 months]

Lifecycle: normal

Key pair assigned at launch: proppem

Monitoring: disabled

Termination protection: Disabled

AMI location: amazon/ubuntu/images/hvm-ssd/ubuntu-jammy-22.04-amd64-server-20230208

Stop-hibernate behavior: Disabled

State transition reason: -

Fig. 3. AWS EC2 instance showing details of the hosted blockchain network.

property information with limited independent sources (Adilieme et al., 2024a). Contact with a client is inevitable, and as highlighted in Klammer et al. (2019), valuers face a 'client contact paradox' that requires sound judgment while balancing interests. Hence, as Klammer et al. (2018) noted, valuers may be able to resist clients' influence due to procedural power, and this can be through the use of blockchain. However, the issue with information power remains as there is still the need for access and verification of property information, which may not be publicly

available.

Furthermore, given that the submission made via the *propchain* was approved by all parties, this cuts out the practice of reviewing and modifying draft reports, which, according to studies like Klammer et al. (2018), is an avenue for clients' influence.

```

19     private String documentLink;
20     @Property()
21     private String clientId;
22     @Property()
23     private String bankId;
24     @Property()
25     private String valuerId;
26     @Property()
27     private String loanId;
28     @Property()
29     private boolean modify;
30     @Property()
31     private boolean askedPermissionForModification;
32
33     @Property()
34     private boolean documentApproved;
35
36     public boolean isDocumentApproved() {
37         return documentApproved;
38     }
39
40     public void setDocumentApproved(boolean documentApproved) {
41         this.documentApproved = documentApproved;
42     }
43
44     public String getDocumentId() {
45         return documentId;
46     }
47
48     public void setDocumentId(String documentId) {
49         this.documentId = documentId;
50     }
51
52     public String getDocumentType() {
53         return documentType;
54     }
55
56     public void setDocumentType(String documentType) {
57         this.documentType = documentType;
58     }
59
60     public String getDocumentTitle() {
61         return documentTitle;
62     }
63
64     public void setDocumentTitle(String documentTitle) {
65         this.documentTitle = documentTitle;
66     }
67
68     public String getDocumentValue() {
69         return documentValue;
70     }
71
72     public void setDocumentValue(String documentValue) {
73         this.documentValue = documentValue;
74     }
75
76     public String getDocumentDescription() {
77         return documentDescription;
78     }

```

Fig. 4. Smart contract for property details and documents.

4.2. Theme 2: ease of use

In investigating the reflections on the potential effectiveness of *propchain*, a theme emerged on the easiness of using a blockchain-based tool for property valuation.

“Limiting my opinion to the app ... You go into the app; you understand what you want to do. You request a valuation, pick a valuer, and that goes through. There aren't any glitches, and you don't need to provide lots of documents; there is no complication in verifying aspects you don't know of. Just request someone you know or do not know, and the app will handle it. Then, the other party gets the notification. So, in-app, everything is straightforward if you understand what you are doing and why

you are doing it. I think most of the issues will be in the backend, ensuring that the app is doing what it's intended to achieve.” (C1).

Participants also alluded that simplicity helped them use the application, which may be a factor in promoting adoption.

“I love it when things are kept short and simple. I just uploaded my valuation report, which I thought was ready ... I also loved that it was so easy to understand that this is exactly what I am doing, and this is my space, so I do not need to bother about the actions of other participants. So long the backend is properly secured, because there could be an issue with the fact that it is quite simple to use, in such a way that there is no interference, there are no possible ways that things could be manipulated. A banker goes to his place and does his work, the valuer goes to where he

```

7*import java.lang.reflect.Type;
26 |
27 @Contract(name = "PropChaincode", info = @Info(title = "Prop Chaincode", description = "Prop chaincode", version = "1.0.0", license = @License(name = "Apache 2.0 License",
28 @Default
29 public final class PropChaincode implements ContractInterface {
30
31     private final Gson gson = new Gson();
32
33     private enum AssetTransferErrors {
34         USER_NOT_FOUND, USER_ALREADY_EXISTS, INVALID_ACCOUNT_TYPE_PROVIDED, INVALID_DATA_PROVIDED, USERNAME_TAKEN,
35         ROLE_NOT_AVAILABLE, SUCCESS
36     }
37
38     /**
39      * Creates some initial assets on the ledger.
40      *
41      * @param ctx the transaction context
42      */
43     @Transaction(intent = Transaction.TYPE.SUBMIT)
44     public void InitLedger(final Context ctx) {
45     }
46
47     /**
48      * Creates a new asset on the ledger.
49      *
50      * @param ctx          the transaction context
51      * @param assetID      the ID of the new asset
52      * @param color         the color of the new asset
53      * @param size          the size for the new asset
54      * @param owner         the owner of the new asset
55      * @param appraisedValue the appraisedValue of the new asset
56      * @return the created asset
57      */
58     @Transaction(intent = Transaction.TYPE.SUBMIT)
59     public User userRegistration(final Context ctx, String userData) {
60         ChaincodeStub stub = ctx.getStub();
61
62         Gson gson = new Gson();
63         User user = gson.fromJson(userData, User.class);
64
65         user.setAccount_type(ctx.getClientIdentity().getAttributeValue("role"));
66
67         checkUserType(user);
68
69         checkUsername(user);
70
71         if (AssetExists(ctx, String.format(Keys.USER_ACCOUNT_INFO_KEY, user.getAccount_type(), user.getUser_name()))) {
72             String errorMessage = String.format("username %s already taken", user.getUser_name());
73             System.out.println(errorMessage);
74             throw new ChaincodeException(errorMessage, AssetTransferErrors.USER_ALREADY_EXISTS.toString());
75         }
76
77         if (AssetExists(ctx, String.format(Keys.USER_ACCOUNT_TYPE_KEY, user.getUser_name()))) {
78             String errorMessage = String.format("username %s already taken", user.getUser_name());
79             System.out.println(errorMessage);
80             throw new ChaincodeException(errorMessage, AssetTransferErrors.USERNAME_TAKEN.toString());
81         }
82

```

Fig. 5. Smart contract for the propchain.

has to do his part, which is uploading the report, and the client also does his part; the ease of use could be a huge motivating factor to adopt it" (PV1).

While the preceding perspectives were positive, the focus group identified and reflected on potential drawbacks. The drawbacks relate to the ease of use, which hinges on simplicity and may not be realistic or practical.

"It was easy to use with direct application. However, I consider this a sample template that does not have some needed details of description, survey, title, maps, basis of valuation, etc., for necessary standard and regulatory monitoring." (PV2).

The drawback was explored in more detail by B2. "When I used the app, it was too simple. At the point of uploading, I only requested a valuer

from the bank's list of valuers; I did not get the opportunity to upload titling documents, which the valuer is to see and use for his valuation. There was also an issue of not having a conversation about the fee. Ideally, the valuer must assess the job, send a bill, or give the bank a fee note. So, these complementary aspects of a property valuation process really need to be investigated. How do we integrate them into the blockchain system while making everything easy to use without being too simple."

Hence, it was acknowledged that whilst ease of use would promote the adoption of new technology, it should not be too simple to the point of doing away with integral parts of the valuation process. Given the findings of Levy and Schuck (2005) on the importance of procedural power and Klammer et al. (2018) on task complexity relating to information and procedure, the proposed solution, blockchain technology, needs to identify the areas of the valuation assignment that needs to be

```

305     transactionHistoryModel2.setTimestamp(ctx.getStub().getTxTimestamp().toString());
306     transactionHistoryModel2.setData(gson.toJson(loan));
307     transactionHistoryModel2.setType("assignedValuerToLoan");
308     transactionHistoryModelsReferee.add(transactionHistoryModel2);
309     stub.putStringState(Keys.LOAN_HISTORY_KEY, new Gson().toJson(transactionHistoryModelsReferee));
310     return loan;
311 }
312
313 /**
314  * Changes the owner of a asset on the ledger.
315  *
316  * @param ctx      the transaction context
317  * @param assetID  the ID of the asset being transferred
318  * @param newOwner the new owner
319  * @return the old owner
320  */
321 @Transaction(intent = Transaction.TYPE.SUBMIT)
322 public ValuerDocument askAdminForPermissionToEditDocument(final Context ctx, String documentId) {
323     if (!ctx.getClientIdentity().assertAttributeValue("role", "client")) {
324         throw new ChaincodeException("You dont have client role",
325             AssetTransferErrors.ROLE_NOT_AVAILABLE.toString());
326     }
327     ChaincodeStub stub = ctx.getStub();
328     Gson gson = new Gson();
329     ValuerDocument valuerDocument = gson.fromJson(stub.getStringState(documentId), ValuerDocument.class);
330
331     try {
332         valuerDocument.setAskedPermissionForModification(true);
333         stub.putStringState(documentId, gson.toJson(valuerDocument));
334         String loanTransactions = stub.getStringState(Keys.LOAN_HISTORY_KEY);
335         ArrayList<TransactionHistoryModel> transactionHistoryModelsReferee = new ArrayList<TransactionHistoryModel>();
336         if (loanTransactions != null && !loanTransactions.isEmpty()) {
337             Type typeOfObjectsList = new TypeToken<ArrayList<TransactionHistoryModel>>() {
338             }.getType();
339             transactionHistoryModelsReferee = new Gson().fromJson(loanTransactions, typeOfObjectsList);
340         }
341
342         TransactionHistoryModel transactionHistoryModel2 = new TransactionHistoryModel();
343         transactionHistoryModel2.setTxId(ctx.getStub().getTxId());
344         transactionHistoryModel2.setTimestamp(ctx.getStub().getTxTimestamp().toString());
345         transactionHistoryModel2.setData(gson.toJson(valuerDocument));
346         transactionHistoryModel2.setType("askedPermissionForEditDocument");
347         transactionHistoryModelsReferee.add(transactionHistoryModel2);
348         stub.putStringState(Keys.LOAN_HISTORY_KEY, new Gson().toJson(transactionHistoryModelsReferee));
349     } catch (Exception e) {
350         e.printStackTrace();
351     }
352     return valuerDocument;
353 }
354
355 /**
356  * Changes the owner of a asset on the ledger.
357  *
358  * @param ctx      the transaction context
359  * @param assetID  the ID of the asset being transferred
360  * @param newOwner the new owner
361  * @return the old owner
362  */
363 @Transaction(intent = Transaction.TYPE.SUBMIT)
364 public ValuerDocument submitDocumentToValuer(final Context ctx, ValuerDocument document) {

```

Fig. 6. Smart contract for the propchain.

integrated into the solution or potentially face non-adoption if it is not fit for purpose.

4.3. Theme 3: Modifying the approach to a valuation assignment

Given the dimensions and persistence of clients' influence in the Nigerian property valuation industry (Adilieme et al., 2024a) and the positive sentiments expressed by the participants towards blockchain solutions, they reflected on how the *propchain* could potentially modify the property valuation exercise. This includes cutting down time spent on some segments of the valuation process, such as verification of records, and creating a store of records for the regulatory body, which creates a version of record that can be referenced in future and introduces a form of assurance due to the element of transparency.

".... it would possibly change the valuation process. For the first part, the bank feels more comfortable knowing there is no client influence on a

valuation report. When we receive a valuation report, we always have to review photos and figures, and sometimes, we have to physically look at the property to be sure it is worth it. So, if we get this thing running live, it reduces the need for a second investigation or an extra inspection. It streamlines everything: you upload the request and documents, a valuer receives it, sends a report, and attaches a bill to it, you look at it, debit the customer, the valuer gets his money, and the bank gets the valuation, and the bank would never be worried about the database So, if a matter were to come up in court, you don't need to start asking the valuer to bring the report to court; the institution should be able to access it from the database, pull it out and know where the exercise went wrong and know how to sanction a valuer as well as any other actions to take." (B1).

"I posit in the affirmative. Firstly, it makes the experience of doing a valuation easier and faster. I recall one of the banks, Diamond Bank, that tried something like this, where through an app, they assign the valuation instruction to a valuer, attach the title documents, and do all of those

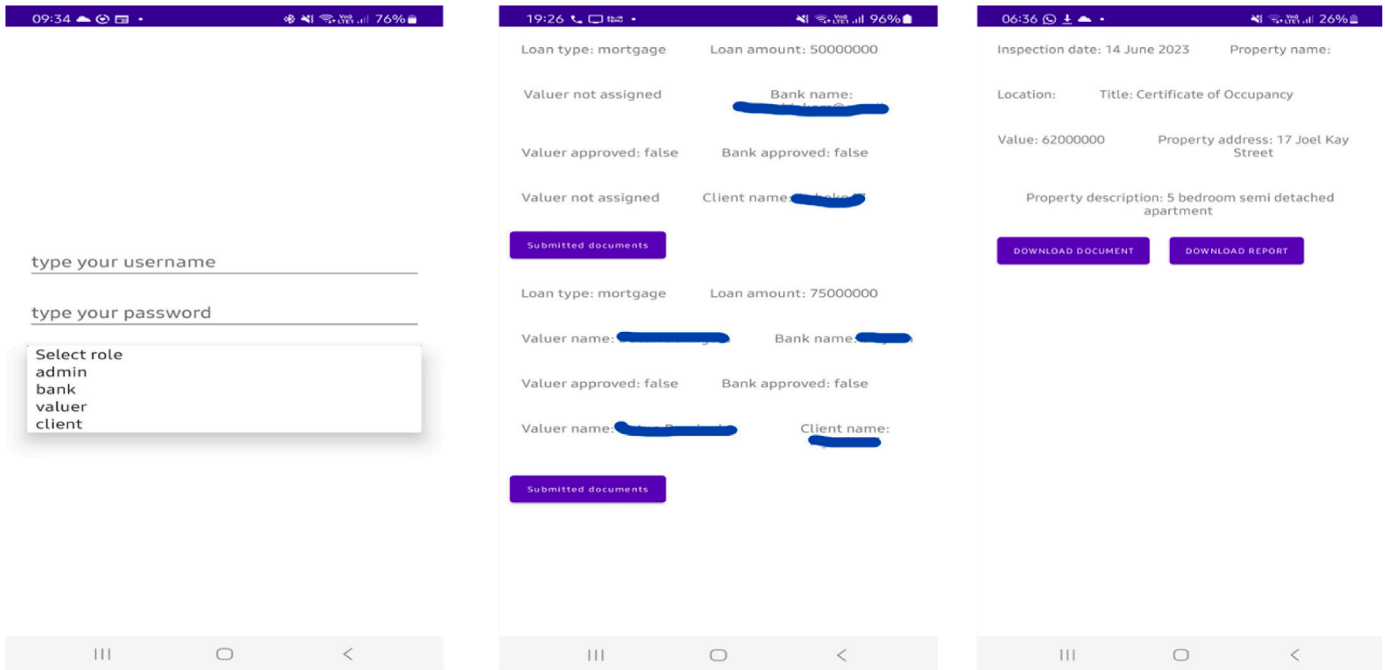


Fig. 7. Screenshots from the propchain in operation.

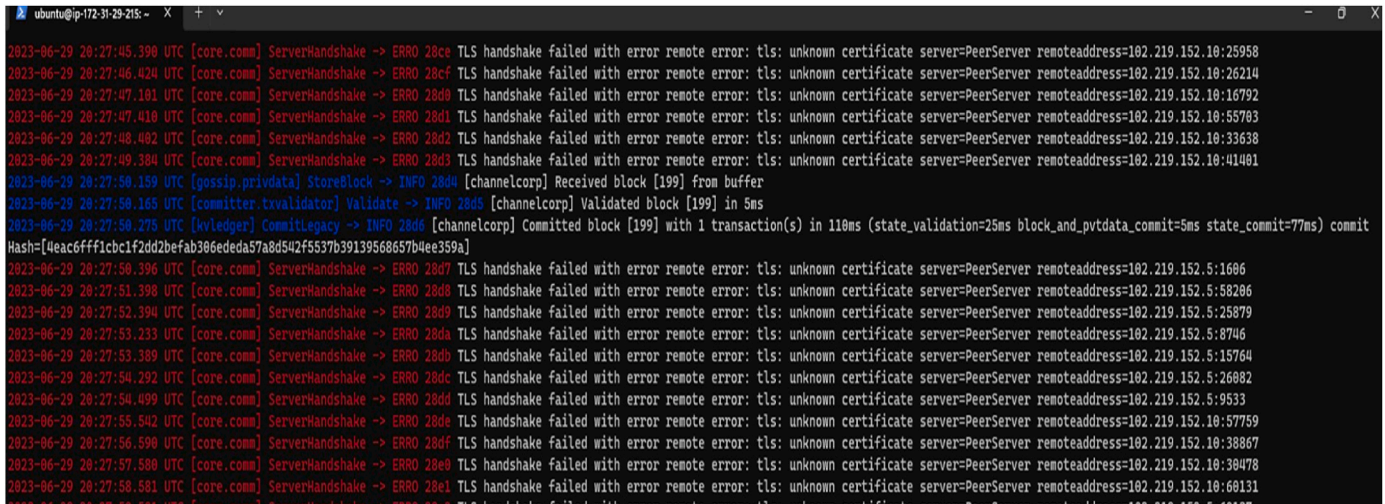


Fig. 8. Record of a validated transaction on the blockchain network.

talks. So, when it pops off on the system that a valuation has been assigned to you, it makes it much easier to see all the title documents and other things you need. Many other banks have technology-enabled systems to speed up the process However, what we have just done would change a lot because, as a valuer, it is easy for me to go in there and upload my report without any challenge and be sure of the element of transparency.” (PV1).

In analysing the participants’ views, the application per se does not modify the approach to a valuation exercise; instead, it brings in a certain level of transparency, information sharing, and certainty and reduces time constraints. These advantages align with the findings of Wang et al. (2020) and Wu et al. (2022) on the ability of a blockchain-based system for construction to achieve information sharing and traceability.

4.4. Theme 4: Issues limiting the effectiveness of the blockchain app in reducing clients’ influence

A theme that emerged was the likely issues and challenges that could impact the effectiveness of using blockchain in the property valuation process to reduce clients’ influence. These include the receptiveness of the stakeholders and regulators and compliance with regulatory guides. These challenges were peculiar to the Nigerian property valuation industry. They may not extend to industries with quality data and better institutional capacity.

“The main challenge I perceive that will affect people’s receptiveness in the property sector to this solution is that you are attempting to introduce an element of detachment where everyone works independently. Because of the backend issues with the land registry and fraudulent valuers, everyone needs a certain level of trust before they agree to deal with valuers or clients. So, you are introducing anonymity, but how do you ensure trust

and that everyone on the app is verified and authenticated? I think these will be the main issues you have to manage. In essence, for people to trust a solution attempting to introduce trust.” (C1).

“The application is useful for facilitating a property valuation process; however, the challenge may be how to process its industry-wide adoption with the involvement of ESVARBON (the regulatory body) and the time it will take. There must be details on how compliance checks with standards and regulatory frameworks will be assessed. Also, the avenue to report or disclose poor delivery should be included. Fourthly, no billing system exists for the valuer or parties to agree on fees. Fifthly, the security of the system may be questioned. It should be foolproof for hacking and other infractions.” (PV2).

A major challenge identified was integrating the blockchain-based application with property databases for seamless data identification, sharing and verification, given that these databases are not readily available in Nigeria.

“for the developed world, there’s a clear database for everything, but we don’t have an existing one. Now, it’s easy for you to say that the banker will stay in his office and send a request to a valuer. The valuer handles it and sends a report. But in this environment, you also know that we tend to have the issue of identifying property. In my line of work, there has been a scenario where people give you an address to go to, and the property does not exist. There needs to be a way to get the required data and databases sorted because it’s not straightforward if you don’t have that part sorted out. The banker must know the site and identify the property and its title ... You can’t log into a Ministry of Lands database to know the ownership and property title. So, automatically, it’s no longer straightforward if you don’t have that information about the titles.” (B1).

PV1 aligned with B1 in their statement, highlighting as follows: “When the instruction is sent to you, many variables are involved. One has to do with the contact with the property owner, just like was rightly articulated, regarding the problem with identifying the property and title. So, it’s not enough to get instruction from the banker; I also need somebody who will take me to this property, allow or give consent for the inspection, and take the necessary physical measurements. You must make physical confirmation visits and measurements even when you locate it on Google or use satellite imagery. We are currently dealing with one in my office where we were asked to value a property. They gave us the survey plan, and from the survey plan, unfortunately, the guy who went for the inspection could not locate the exact beacons. He just assumed that the property was the plot, and now it is a legal matter because the owners are saying they want to sell the property, but the one valued is not the one they asked us to value.” (PV1).

In discussing challenges, PV1 elaborated on the issue of access to property for valuation assignments. This is a hurdle that blockchain technology may not be able to scale within the Nigerian context as property owners wield much power in giving access to property and specific data and can leverage that to their advantage since it is an off-chain action.

“Don’t forget that in some instances, the property owner never mentioned to the occupiers that the property is to be used for collateral and that valuers will be coming for inspection. Hence, I think that beyond sending the report on the app, many issues must be considered, including who takes you to inspect the property and how you take the physical measurement. I can also tell you we have an instance where a bank has given us valuation instruction, and when you are trying to take pictures, people come around, and if you are not careful, you are being mobbed. In some instances, I have done many jobs for some banks in this part of the world, and we had to tell them to come with us to the site. We must take their picture with the property to guarantee our safety.” (PV1).

The challenges identified pointed to human and environmental issues peculiar to the Nigerian environment that a technological intervention cannot address without a robust regulatory environment and culture of information sharing. The absence of support databases and

data has been attributed to confidentiality, lack of cooperation, and data accuracy, among other factors (Olapade & Olaleye, 2019). Nevertheless, given the desire by valuers in Nigeria to share information and trust databases managed by their professional body (Olapade et al., 2019; Olapade & Olaleye, 2018), a potential gain from the *propchain* is the storage of data which could be used to develop a valuation-based property database.

On the challenge raised with valuer safety and stakeholder receptiveness to the technology, which may potentially expose below-par practices, our findings align with the position of Konashevych (2020) on the potential for blockchain to be inhibited by political will and corruption as opposed to issues around the technology. Citing evidence from Honduras, Konashevych (2020) highlighted that the stumbling block is usually due to a lack of institutional support, as blockchain technology still depends on political will for successful adoption to promote property administration. Furthermore, Yapicioglu and Leshinsky (2020) argue that for a blockchain to be implemented, there must be legal and governance structures to support its success. Hence, this presents a conundrum given the rating of the Nigerian property market, which is ranked a low, transparent country - 64th out of 89 global property markets (JLL, 2024). Furthermore, the corruption perceptions index is relatively poor, ranking 145 out of 180 countries in 2023 (Transparency International, 2024). Given the findings in Adilieme et al. (2024a) on the pervasiveness of clients’ influence and the use of economic power and unlevel structure between institutional clients and valuers, the governance structures may be unable to support blockchain integration.

4.5. Theme 5: Mechanism for promoting adoption

The perspectives expressed highlighted that technological interventions are not entirely new. Hence, the mechanisms to promote the adoption of a blockchain solution in property valuation include clear communication with stakeholders to educate them on the benefits, securing the buy-in of large organisations and the government, and proactive action by the regulatory body and professional association.

“The clients’ influence considerations you are making is an existing issue. Essentially, some people, like valuers, banks, and clients, make bread and butter from client influence existing as an issue. So, you are bringing in an app intended to solve the problem, essentially putting anonymous reviewers on their backs. You are attempting to get them to operate on your platform, forgetting that they might think they could get in trouble for a value or a mistake or a result of clients’ pressure, which they did in the past or on a valuation, be it intentional or not. I think it will be good first to make them understand the gains and independence that come from using blockchain and then make the database accessible to the registry of valuers or even to everyone ... Again, considering Africa and Nigeria, there will still be significant pushback from anyone who benefits from client influence being a prevalent issue.” (C1).

“One way will be to get stakeholders on board. The banks will jump at it on one end since they have always tried to apply technology tools to deliver valuations faster and more transparently. Big organisations in other sectors that periodically require valuations would jump at it if the idea and objective were communicated to them. So, while it may not be something that everybody is going to jump on board at the same time. Some people gain trust after a while, and it becomes generally accepted. But once you start with the banks and government parastatals, it will go out there because it already solves an existing problem for us. It improves the interface between valuers and banks, who I think are the largest clients of valuers with regard to valuation. The valuation process will become more trusted when the government streamlines a lot based on regulation.” (B1).

“I can’t agree more with the submissions. I mention most of the time that from my end, as a partner in a nationally large valuation practice, with

the type of jobs we get, 85 to 90% of the valuations that are being done annually in Nigeria are mortgage-based and being used by the financial institutions. I think it's something that will be gladly accepted. Currently, the banks keep introducing technology tools, so this will be another tool, but we will have to establish clear communication and ensure that the association and regulators are on board by providing regulations that will support this tool and enforce punitive measures for members who are identified as still erring. This will ensure that people see merit in a transparent system. I don't want to think we can eliminate client influence, especially considering what was said earlier on the other issues with databases. But I think it will eliminate the influence of clients, and it will be accepted. I personally look forward to it." (PV1).

As highlighted in the previous theme, the study of [Konashevych \(2020\)](#) and [Yapicioglu and Leshinsky \(2020\)](#) shows that blockchain adoption requires political will and institutional support. The participants' reflections echo this. Hence, proactive action will be needed to build up regulatory systems and get support from the government to promote this tool for transparency.

5. Discussions

Given that one of the primary goals of blockchain is to ensure information exchange among non-trusting parties ([Werbach, 2018](#)), the findings indicate the huge possibility of that goal within the property valuation space. Nevertheless, the challenges raised in the findings are consistent with the issues raised by [Konashevych \(2020\)](#), [Yapicioglu and Leshinsky \(2020\)](#), and [Konashevych \(2021\)](#) on the ability of blockchain to play the role of a trusted intermediary where there is no political or institutional support. Accordingly, this presents a paradox for this study; Nigeria, from an institutional front, is fraught with poor property market transparency, reported cases of corruption, and worsening macroeconomic situations ([JLL, 2024](#); [Transparency International, 2024](#); [Focus Economics, 2024](#); [Central Bank of Nigeria, 2024](#)). Furthermore, the Nigerian property valuation industry does not show the presence of sufficient institutional capacity ([Adilieme et al., 2024a](#)). Hence, whilst the technology holds promise and shows its output as an intermediary for information transparency, the challenges and gaps revealed in the findings relating to data, safety, and stakeholder trust threaten its effectiveness and need to be considered by stakeholders.

[Fig. 9](#) visualises a potential gap identified during the focus group discussion based on Nigeria's current valuation process and system. Blockchain technology efficiently facilitates communication during the instruction, analysis, and reporting phases, and all activities are stored and visible on-chain. However, due to the issue of databases, access and institutional framework, which makes obtaining data and information without client substantial input difficult, the inspection phase provides an opportunity for clients influence through information power and a mix of reward power as highlighted in studies on clients influence in Nigeria (see [Amidu et al., 2007a](#); [Nwuba et al., 2015a](#); [Adilieme et al., 2024a](#)).

Consequently, whilst [Freybote et al. \(2014\)](#) provide evidence on the efficacy of information transparency as promoted with legal tools, in this case, technology delivers on being a conduit for such transparency to address the issue of clients' influence. However, the challenges presented to its operations in an institutionally weak environment are acknowledged. Hence, this study recommends that the regulatory capacity and issue of data be resolved first to allow the blockchain technology to integrate and function seamlessly.

The importance of building institutional capacity and improving the regulatory system is not far-fetched. For instance, [Mizen \(2024\)](#), using the audit industry and fallout from the information power manipulation by a Big Four professional firm, PwC in Australia, reports that the corporate regulator, the Australian Securities and Investments Commission, overhauled its oversight function and approach as it was at risk of being captured by the same companies it was supposed to regulate. This is of particular importance to Nigeria, given the issues identified in the property valuation industry in Nigeria about the ineffective regulatory framework and system, which has led to problems such as clients' influence and professional services underpricing, among others ([Adilieme et al., 2024a](#); [Babawale, 2013](#); [Oladokun & Mooya, 2020](#)). Hence, much will have to be done to support the proposed solution.

Theoretically, the findings question the ubiquitous affirmation around the effectiveness of blockchain technology as a tool for transparency and align with [Konashevych \(2020\)](#), [Yapicioglu and Leshinsky \(2020\)](#), and [Konashevych \(2021\)](#) on the technology needing institutional support. Hence, it cannot completely replace traditional systems if those supports are not in place. This raises the question of whether blockchain technology can be effectively implemented in places like Nigeria, which needs it since the problems it will solve exist due to

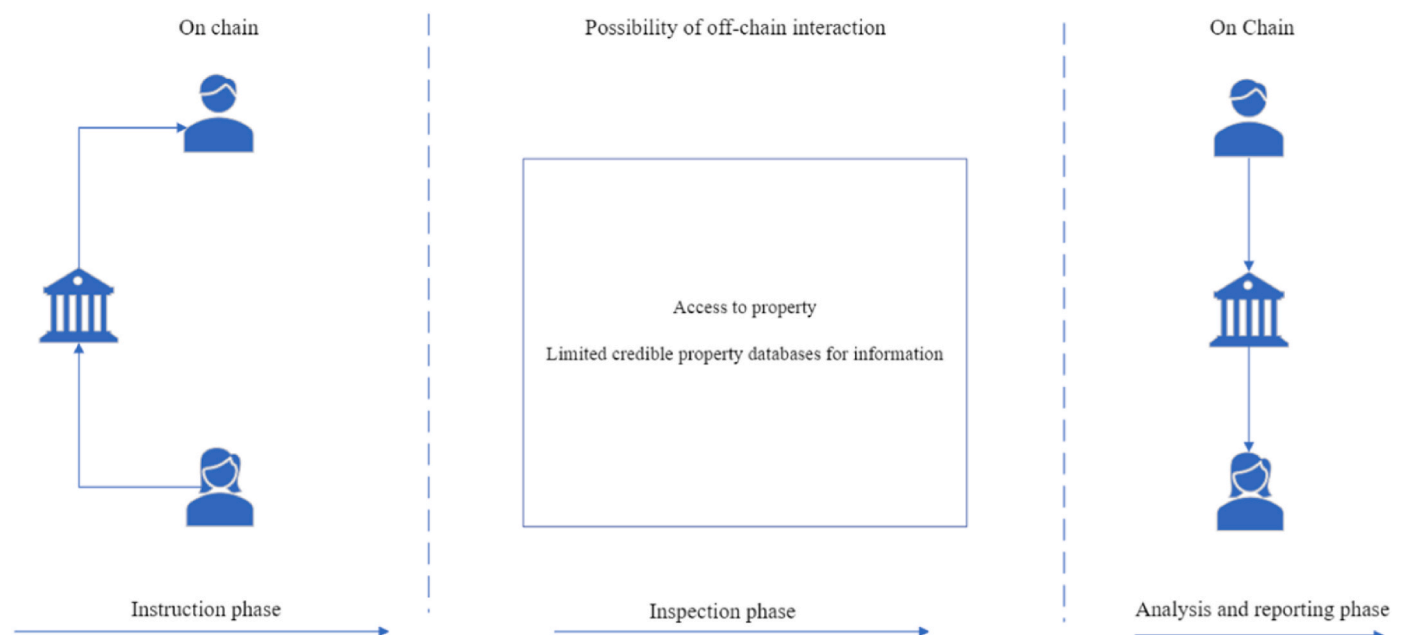


Fig. 9. Potential gap in the application of blockchain technology.

inefficient institutions. Nevertheless, blockchain presents a viable use case to increase transparency and reduce valuation time and risk to property valuers in developed, mature, and transparent property markets. For instance, a white paper by the [Hong Kong Monetary Authority \(2016\)](#) reported that blockchain technology has been successfully trialled in Hong Kong to support the mortgage application process with the property valuation component included in this process. According to the report, the technology reportedly reduced the application process time, digitalised the mortgage application and property valuation component and facilitated seamless communication, information sharing and verification between the bank and valuers. Whilst no further studies have affirmed the findings and submissions made because of the tested proof of concept in Hong Kong, which included property valuation as part of a mortgage process framework, this study acts as proof of the place of blockchain as a tool which can be incorporated into the property valuation process for communication and information management.

6. Conclusions

Clients' influence, an acknowledged issue in property valuation, is a recurrent problem for untransparent and developing property markets. Proposed solutions include legal and regulatory interventions, valuer integrity, ethics, and information transparency. However, evidence from the literature suggests that the problem of client influence persists in countries such as Nigeria. Hence, this study proposes and explores the use of blockchain technology to address the clients' influence challenge in the property valuation industry. To achieve this goal, an application, *Propchain*, was developed on the Hyperledger Fabric Network to deliver information transparency and was tested by valuers, bankers, and clients. The findings indicate a huge possibility for the integration of blockchain for property valuation, with the potential to reduce the influence of clients.

While the findings show prospects for reducing clients' influence through blockchain technology, there was no strong evidence that using such technology would totally eliminate clients' influence in Nigeria due to systemic gaps in the valuation industry, which would limit its effectiveness and allow backdoor channels for opportunistic clients. This includes an absence of databases, which still makes clients in Nigeria the significant sources of information and prevents seamless verification of property information. Furthermore, issues of legal support and stakeholder trust were barriers that the *propchain* could not address. The implications of this study include the following: Firstly, blockchain as an emerging proptech can be of use within property valuation. Secondly, for blockchain to be effectively applied in property valuation, institutional support and stakeholder trust are required, which may, in essence, call into question the impact of blockchain since the idea is to introduce transparency and trust and act as a supplementary tool for a trustless environment. Finally, the themes from this study are useful for developing conceptual frameworks that cover elements that facilitate and inhibit the development of blockchain-based tools and solutions applicable to property valuation practice.

Consequently, to promote blockchain's effectiveness in reducing client influence, this study recommends that support systems such as property databases, the legal and regulatory framework, and monitoring systems be strengthened to minimise the flaws in the designed system and increase the potential for information transparency in developing markets. Our study highlights that blockchain technology holds significant promise for enhancing transparency and reducing client influence in property valuation processes. This innovative solution can be effectively integrated into valuation guidelines, particularly in developed markets with established institutions and information transparency. Global valuation professional organisations, such as the Royal Institution of Chartered Surveyors (RICS), could consider adopting blockchain technology as part of their valuation practices. Following the successful proof of concept trialled in Hong Kong, blockchain could serve as an intermediary for communication and information sharing in developed

markets. This would build on existing information transparency frameworks and institutional support, further enhancing the reliability and efficiency of property valuations.

Additionally, blockchain's role could be complemented by integrating other technological advancements, such as machine learning and artificial intelligence, to provide a comprehensive approach to valuation practices. Although client influence is often a focal point in developing markets, evidence from the audit industry suggests that it can still be a concern in developed markets. Therefore, blockchain technology has the potential to address these challenges by improving the overall integrity and accuracy of the valuation process.

The research has some limitations, which provide potential areas for future research. Firstly, the blockchain network was not technically evaluated for performance and scalability. Hence, it is a simple solution to facilitate an empirical study, and real-life cases may require a solution with more features. Furthermore, the valuation exercise was done using pseudo cases and dummy data. Thirdly, the single round focus group saw important perspectives and themes emerge; nevertheless, further rounds of focus group may have provided more perspectives. Finally, some participants may be susceptible to withholding sensitive information in group settings, potentially influencing the interpretation of the findings; nevertheless, as highlighted earlier, the participants had established contact with the research through interviews and were familiar with the project and put at ease.²

Despite these limitations, the study's findings are valuable for property valuation researchers and practitioners in understanding the potential for blockchain and how to develop a solution with blockchain technology. Future research can evaluate the performance of applications built on the Hyperledger Fabric network and use real-life property valuation exercises to assess actual scenarios and data. Furthermore, different methods can be used to assess the participants' perceptions.

CRediT authorship contribution statement

Chibuikem Michael Adilieme: Writing – original draft, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Rotimi Boluwatife Abidoye:** Writing – review & editing, Supervision. **Chyi Lin Lee:** Writing – review & editing, Supervision.

Declaration of competing interest

No competing interest to declare.

Acknowledgements

1. This paper forms part of a doctoral research on reducing clients' influence in property valuation, and the first author has been supported by a University Postgraduate Award scholarship for Higher Degree Research Students by UNSW Sydney.
2. The acknowledges Dr Samuel Yaw Frimpong for initially reading and providing suggestions to the manuscript.
3. The authors thank the anonymous referees for the comments and suggestions made to improve the quality of the paper.

References

- Achu, K., Wan Chin, L., Burhan, B., & Nordin, M. F. (2015). Factors affecting client influence on property valuation in Malaysia: Do client size and size of value adjustment matter? *Jurnal Teknologi*, 75(10), 39–49. <https://doi.org/10.11113/jt.v75.5271>

² In line with the ethical requirements, participants were provided information sheets containing contact information about the research team and ethics coordinator if they were uncomfortable discussing anything.

- Adilieme, C. M., Abidoye, R. B., & Lee, C. L. (2023). Client influence in property valuation: A scoping literature review. *Property Management*, 41(5), 629–646. <https://doi.org/10.1108/pm-09-2022-0061>
- Adilieme, C. M., Abidoye, R. B., & Lee, C. L. (2024a). Client influence in property valuation in Nigeria: A valuer-banker perspective and fuzzy DEMATEL study. *Journal of Property Investment & Finance*, 42(3), 223–246. <https://doi.org/10.1108/jpif-08-2023-0077>
- Adilieme, C. M., Abidoye, R. B., & Lee, C. L. (2024b). Blockchain in property valuation: Perspectives of property professionals in Nigeria. *30th annual pacific rim real estate society conference proceedings*. January 14 to 17, 2024 <https://www.prrs.org/uploads/1352/241/Blockchain-in-Property-Valuation-Perspectives-of-Property-Professionals-in-Nigeria.docx>
- Adilieme, C. M., Abidoye, R. B., & Lee, C. L. (2024c). Barriers and prospects for the adoption of blockchain technology in property valuation. *Journal of European Real Estate Research*. <https://doi.org/10.1108/jer-04-2024-0022>
- Ahmad, R. W., Hasan, H., Jayaraman, R., Salah, K., & Omar, M. (2021). Blockchain applications and architectures for port operations and logistics management. *Research in Transportation Business & Management*, 41, Article 100620. <https://doi.org/10.1016/j.rtbm.2021.100620>
- Ameyaw, P. D., & de Vries, W. T. (2021). Toward smart land management: Land acquisition and the associated challenges in Ghana. A look into a blockchain digital land registry for prospects. *Land*, 10(3), 239. <https://doi.org/10.3390/land10030239>
- Amidu, A., & Aluko, B. T. (2007a). Client influence on valuation: Perceptual analysis of the driving factors. *International Journal of Strategic Property Management*, 11(2), 77–89. <https://doi.org/10.3846/1648715X.2007.9637562>
- Amidu, A., & Aluko, B. T. (2007b). Client influence in residential property valuations: An empirical study. *Property Management*, 25(5), 447–461. <https://doi.org/10.1108/02637470710824720>
- Ansenberg, U., Avni, N., & Rosen, G. (2024). Exploring real estate valuation practices in an informal market. *Habitat International*, 147, Article 103069. <https://doi.org/10.1016/j.habitatint.2024.103069>
- Augustyniak, H., Laszek, J., Olszewski, K., & Waszczuk, J. (2018). Property valuation for mortgage purposes in Poland. *Property Management*, 36(2), 234–247. <https://doi.org/10.1108/pm-09-2016-0049>
- Babawale, G. K. (2013). Valuation accuracy – the myth, expectation and reality. *African Journal of Economic and Management Studies*, 4(3), 387–406. <https://doi.org/10.1108/AJEMS-11-2011-0106>
- Baker, H. K., Nikbakht, E., & Smith, S. S. (2021). Blockchain: An overview. In H. K. Baker, E. Nikbakht, & S. S. Smith (Eds.), *The emerald handbook of blockchain for business* (pp. 3–14). Bingley: Emerald Publishing Limited. <https://doi.org/10.1108/978-1-83982-198-120211003>
- Bellman, L. (2018). High-impact information types on market value: Property appraisers' information sources and assessment confidence. *Journal of Property Research*, 35(2), 139–163. <https://doi.org/10.1080/09599916.2018.1443152>
- Bhatia, S., Douglas, E. K., & Most, M. (2020). Blockchain and records management: Disruptive force or new approach? *Records Management Journal*, 30(3), 277–286. <https://doi.org/10.1108/RMJ-08-2019-0040>
- Braun, V., & Clarke, V. (2006). Using thematic analysis in psychology. *Qualitative Research in Psychology*, 3(2), 77–101. <https://doi.org/10.1191/1478088706qp0630a>
- Central Bank of Nigeria. (2024). Inflation rate. available at <https://www.cbn.gov.ng/rates/infrates.asp>. (Accessed 11 April 2024).
- Cheloti, I., & Mooya, M. (2021). Valuation problems in developing countries: A new perspective. *Land*, 10(12), 1352. <https://doi.org/10.3390/land10121352>
- Cheloti, I., & Mooya, M. (2023). Property valuation problems and market context – evidence from Kenya. *Journal of Property Research*, 40(1), 76–100. <https://doi.org/10.1080/09599916.2022.2119879>
- Chen, F. Y., & Yu, S. M. (2009). Client influence on valuation: Does language matter? *Journal of Property Investment & Finance*, 27(1), 25–41. <https://doi.org/10.1108/14635780910926658>
- Cosmos. (2023). Cosmos features. <https://cosmos.network/features>
- Crosby, N., & Devaney, S. (2019). Appraisal-based indices. In B. D. MacGregor, R. Schulz, & R. K. Green (Eds.), *Routledge companion to real estate investment* (1st ed., pp. 172–191). Routledge.
- Crosby, N., Devaney, S., Lizieri, C., & Mcallister, P. (2018). Can institutional investors bias real estate portfolio appraisals? Evidence from the market downturn. *Journal of Business Ethics*, 147(3), 651–667. <https://doi.org/10.1007/s10551-015-2953-1>
- Crosby, N., Hutchison, N., Lusht, K., & Yu, S. M. (2019). Valuations and their importance for real estate investments. In B. D. MacGregor, R. Schulz, & R. K. Green (Eds.), *Routledge companion to real estate investment* (1st ed., pp. 143–171). Routledge.
- Davis, M., Lennerfors, T. T., & Tolstoy, D. (2022). Can blockchain-technology fight corruption in MNEs' operations in emerging markets? *Review of International Business and Strategy*, 32(1), 39–56. <https://doi.org/10.1108/ribs-12-2020-0155>
- Ding, L., & Nakamura, L. (2016). The impact of the Home valuation code of conduct on appraisal and mortgage outcomes. *Real Estate Economics*, 44(3), 658–690. <https://doi.org/10.1111/1540-6229.12120>
- Focus Economics. (2024). Nigeria interest rate. available at <https://www.focus-economics.com/country-indicator/nigeria/interest-rate/#:~:text=At%20its%2024%E2%80%939325%20July,to%2018.75%25%20from%2018.50%25>. (Accessed 11 April 2024).
- Freybote, J., Ziobrowski, A., & Gallimore, P. (2014). Residential real estate appraisal bias in the absence of client feedback. *Journal of Housing Research*, 23(2), 127–142. <https://doi.org/10.1080/10835547.2014.12092091>
- Gilbertson, B., & Preston, D. (2005). A vision for valuation. *Journal of Property Investment & Finance*, 23(2), 123–140. <https://doi.org/10.1108/14635780510699998>
- Graglia, J. M., & Mellon, C. (2018). Blockchain and property in 2018: At the end of the beginning. *Innovations: Technology, Governance, Globalization*, 12(1–2), 90–116. <https://doi.org/10.1162/inov.a.00270>
- Grant, D., & Cherif, E. (2016). Using design science to improve web search innovation in real estate. *Journal of Organizational Computing & Electronic Commerce*, 26(3), 267–284. <https://doi.org/10.1080/10919392.2016.1194064>
- Guest, G., Namey, E., Taylor, J., Eley, N., & McKenna, K. (2017). Comparing focus groups and individual interviews: Findings from a randomized study. *International Journal of Social Research Methodology*, 20(6), 693–708. <https://doi.org/10.1080/13645579.2017.1281601>
- Hamedari, H., & Fischer, M. (2021). Construction payment automation using blockchain-enabled smart contracts and robotic reality capture technologies. *Automation in Construction*, 132, Article 103926. <https://doi.org/10.1016/j.autcon.2021.103926>
- Hijazi, A. A., Perera, S., Calheiros, R. N., & Alashwal, A. (2023). A data model for integrating BIM and blockchain to enable a single source of truth for the construction supply chain data delivery. *Engineering Construction and Architectural Management*, 30(10), 4645–4664. <https://doi.org/10.1108/ecam-03-2022-0209>
- Holt, R., Hoelscher, J., Hoskins, B., & McCann, J. (2015). Dodd-Frank impact on mortgage appraisal process. *European Journal of Business and Social Sciences*, 3(12), 58–67.
- Hong Kong Monetary Authority. (2016). Whitepaper on distributed ledger technology. https://www.hkma.gov.hk/media/eng/doc/key-functions/financial-infrastructure/Whitepaper_On_Distributed_Ledger_Technology.pdf
- Hooper, A., & Holtbrügge, D. (2020). Blockchain technology in international business: Changing the agenda for global governance. *Review of International Business and Strategy*, 30(2), 183–200. <https://doi.org/10.1108/ribs-06-2019-0078>
- Iban, E. C. (2022). An explainable model for the mass appraisal of residences: The application of tree-based Machine Learning algorithms and interpretation of value determinants. *Habitat International*, 1128, Article 102660. <https://doi.org/10.1016/j.habitatint.2022.102660>
- Jafari, P., Shojaei, D., Rajabifard, A., & Ngo, T. (2024). Automating property valuation at the macro scale of suburban level: A multi-step method based on spatial imputation techniques, machine learning and deep learning. *Habitat International*, 148, Article 103075. <https://doi.org/10.1016/j.habitatint.2024.103075>
- JLL. (2024). Global real estate transparency index, 2024. available at: <https://www.jll.com.au/en/trends-and-insights/research/global-real-estate-transparency-index>. (Accessed 3 July 2022).
- Kinnard, W. N., Lenk, M. M., & Worzala, E. M. (1997). Client pressure in the commercial appraisal industry: How prevalent is it? *Journal of Property Valuation and Investment*, 15(3), 233–244. <https://doi.org/10.1108/14635789710184952>
- Klamer, P., Bakker, C., & Gruis, V. (2018). Complexity in valuation practice: An inquiry into valuers' perceptions of task complexity in the Dutch real estate market. *Journal of Property Research*, 35(3), 209–233. <https://doi.org/10.1080/09599916.2018.1510429>
- Klamer, P., Grius, V., & Bakker, C. (2019). How client attachment affects information verification in commercial valuation practice. *Journal of Property Investment & Finance*, 37(6), 541–554. <https://doi.org/10.1108/JPIF-04-2019-0059>
- Klamer, P., Grius, V., & Bakker, C. (2021). The ideal type of valuer: Expert, service provider or reporter? An investigation into prevailing role types in commercial valuation. *Property Management*, 39(2), 210–226. <https://doi.org/10.1108/pm-03-2020-0019>
- Konashevych, O. (2020). Constraints and benefits of the blockchain use for real estate and property rights. *Journal of Property, Planning and Environmental Law*, 12(2), 109–127. <https://doi.org/10.1108/jppel-12-2019-0061>
- Konashevych, O. (2021). 'GoLand registry' case study. *D.G.O 2021: The 22nd annual international conference on digital government research*. <https://doi.org/10.1145/3463677.3463720>
- Kucharska-Stasiak, E., Żróbek, S., & Cellmer, R. (2018). Forms and effectiveness of the client's influence on the market value of property - case study. *Real Estate Management and Valuation*, 26(3), 82–92. <https://doi.org/10.2478/remav-2018-0027>
- Ledger Insights. (2019). Cosmos, a new public-private network of blockchains goes live. <https://www.ledgerinsights.com/cosmos-blockchain-public-private-network-live/>
- Lee, C. L., & Lee, M.-L. (2014). Do European real estate stocks hedge inflation? Evidence from developed and emerging markets. *International Journal of Strategic Property Management*, 18(2), 178–197. <https://doi.org/10.3846/1648715x.2014.925521>
- Lee, M. H., Peng, C. W., & Liao, H. F. (2020). An analysis of objectivity in the real estate appraisal process. *International Real Estate Review*, 23(4), 1109–1130.
- Leite, S. G., Albuquerque, B. A., & Pinheiro, R. P. (2021). Applications of technological solutions in primary ways of preventing transmission of respiratory infectious diseases - a systematic literature review. *Public Health*, 18(20), Article 10765. <https://doi.org/10.3390/ijerph182010765>
- Levy, D., & Schuck, E. (2005). The influence of clients on valuations: The clients' perspective. *Journal of Property Investment & Finance*, 23(2), 182–201. <https://doi.org/10.1108/14635780510584364>
- Liao, H. F., Chu, N. Y., & Peng, C. W. (2018). Awareness of independence of real estate appraisers: An empirical analysis. *International Real Estate Review*, 21(3), 295–316.
- Liu, N., Duncan, R., & Chapman, A. (2020). A critical review of distributed ledger technology and its applications in real estate. Royal Institution of Chartered Surveyors. <https://www.rics.org/globalassets/rics-website/media/knowledge/research/research-reports/rics0077-001-distributed-ledger-technology-review-report-final.pdf>
- Maphosa, V. (2022). Promoting access to COVID-19 Information by underserved communities through the development of a mHealth app. *Cogent Public Health*, 9(1). <https://doi.org/10.1080/27707571.2022.2095722>

- Mashatan, A., Lemieux, V., Lee, S. H., Szufel, P., & Roberts, Z. (2021). Usurping double-ending fraud in real estate transactions via blockchain technology. *Journal of Database Management*, 32(1), 27–48. <https://doi.org/10.4018/JDM.2021010102>
- McAllister, P. (2019). Fund management and governance. In B. D. MacGregor, R. Schulz, & R. K. Green (Eds.), *Routledge companion to real estate investment* (1st ed., pp. 318–337). Routledge.
- Mizen, R. (2024). 'Risk of regulatory capture': ASIC hits back at audit criticisms. *Financial Review*. <https://www.afr.com/politics/federal/risk-of-regulatory-capture-asic-hits-back-at-audit-criticisms-20240416-p5fk4v/>.
- Nanayakkara, S., Rodrigo, M. N. N., Perera, S., Weerasuriya, G. T., & Hijazi, A. A. (2021). A methodology for selection of a Blockchain platform to develop an enterprise system. *Journal of Industrial Information Integration*, 23, Article 100215. <https://doi.org/10.1016/j.jii.2021.100215>
- Newell, G., Muhammad Najib, B. M. R., & Martin Juanil, D. (2010). Assessing client perceptions of the quality of commercial valuation reports in Malaysia. *Pacific Rim Property Research Journal*, 16(4), 458–476. <https://doi.org/10.1080/14445921.2010.11104314>
- Nwuba, C. C., Egwuatu, U. S., & Salawu, B. M. (2015a). Clients' means of influence on mortgage valuation in Nigeria. *Property Management*, 33(4), 367–385. <https://doi.org/10.1108/pm-03-2014-0016>
- Nwuba, C. C., Egwuatu, U. S., & Salawu, B. M. (2015b). Client influence on valuation: Valuers' motives to succumb. *Journal of Property Research*, 32(2), 147–172. <https://doi.org/10.1080/09599916.2015.1005117>
- Oladokun, S. O., & Mooya, M. M. (2020). Pricing of property valuation services in lagos Nigeria: The systems and the effects on valuation practice. *Property Management*, 38(1), 82–108. <https://doi.org/10.1108/PM-04-2019-0024>
- Olapade, D. T., Ekemode, B. G., & Olaleye, A. (2019). Considerations for the design and management of property database in opaque markets. *Journal of Property Investment & Finance*, 37(5), 445–454. <https://doi.org/10.1108/jpiif-10-2018-0080>
- Olapade, D. T., & Olaleye, A. (2018). Resolving the data debacle in commercial property. *Journal of Property Investment & Finance*, 36(3), 295–304. <https://doi.org/10.1108/jpiif-07-2017-0054>
- Olapade, D. T., & Olaleye, A. (2019). Factors affecting accessibility to property data in an opaque market. *Property Management*, 37(1), 82–96. <https://doi.org/10.1108/pm-01-2017-0004>
- Orji, I. J., Kusi-Sarpong, S., Huang, S., & Vazquez-Brust, D. (2020). Evaluating the factors that influence blockchain adoption in the freight logistics industry. *Transportation Research Part E*, 141, Article 102025. <https://doi.org/10.1016/j.tre.2020.102025>
- Parker, A., & Titter, J. (2006). Focus group method and methodology: Current practice and recent debate. *International Journal of Research and Method in Education*, 29(1), 23–37. <https://doi.org/10.1080/01406720500537304>
- Peffer, K., Tuunanen, T., Rothenberger, M. A., & Chatterjee, S. (2007). A design science research methodology for information systems research. *Journal of Management Information Systems*, 24(3), 45–77. <https://doi.org/10.2753/mis0742-1222240302>
- Perera, S., Hijazi, A. A., Weerasuriya, G. T., Nanayakkara, S., & Rodrigo, M. N. N. (2021). Blockchain-based trusted property transactions in the built environment: Development of an incubation-ready prototype. *Buildings*, 11, 560. <https://doi.org/10.3390/buildings11110560>
- Pongnumkul, S., Khonnasee, C., Lertpattanasak, S., & Polprasert, C. (2020). Proof-of-Concept (PoC) of land mortgaging process in blockchain-based land registration system of Thailand. *2nd international conference on blockchain technology (ICBCT 2020)*. <https://doi.org/10.1145/3390566.3391669>
- Pradeep, A. S. E., Yiu, T. W., Zou, Y., & Amor, R. (2021). Blockchain-aided information exchange records for design liability control and improved security. *Automation in Construction*, 126, Article 103667. <https://doi.org/10.1016/j.autcon.2021.103667>
- Rudolph, P. M. (1998). Will mandatory licensing and standards raise the quality of real estate appraisals? Some insights from agency theory. *Journal of Housing Economics*, 7(2), 165–179. <https://doi.org/10.1006/jhec.1998.0226>
- Sanderford, A. R., Read, D. C., Xu, W., & Boyle, K. J. (2017). Obtaining differentiation premiums in the presence of fee regulation in the residential real estate appraisal industry. *Housing Policy Debate*, 27(5), 698–711. <https://doi.org/10.1080/10511482.2017.1305979>
- Sasu, D. D. (2023). *Distribution of devices used for internet access in Nigeria in 2020*. Statista. <https://www.statista.com/statistics/1139297/distribution-of-devices-used-for-internet-access-in-nigeria/#:~:text=In%202020%2C%20mobile%20devices%20were,by%20use%20of%20a%20desktop.>
- Shang, Q., & Price, A. (2019). A blockchain-based land titling project in the republic of Georgia: Rebuilding public trust and lessons for future pilot projects. *Innovations: Technology, Governance, Globalization*, 12(3–4), 72–78. https://doi.org/10.1162/inov_a_00276
- Smithson, J. (2000). Using and analysing focus groups: Limitations and possibilities. *International Journal of Social Research Methodology*, 3(2), 103–119. <https://doi.org/10.1080/136455700405172>
- Smolen, G. E., & Hambleton, D. C. (1997). Is the real estate appraiser's role too much to expect? *The Appraisal Journal*, 65(1), 9–17.
- Soltani, A., & Lee, C. L. (2024). The non-linear dynamics of south Australian regional housing markets: A machine learning approach. *Applied Geography*, 166, Article 103248. <https://doi.org/10.1016/j.apgeog.2024.103248>
- Statista. (2024). Real estate. available at <https://www.statista.com/outlook/fmo/real-estate/nigeria>. (Accessed 17 May 2024).
- Taecharungroj, V. (2021). Google Maps amenities and condominium prices: Investigating the effects and relationships using machine learning. *Habitat International*, 118, Article 102463. <https://doi.org/10.1016/j.habitatint.2021.102463>
- Thackway, W., Ng, M., Lee, C. L., & Pettit, C. (2023a). Building a predictive machine learning model of gentrification in Sydney. *Cities*, 134, Article 104192. <https://doi.org/10.1016/j.cities.2023.104192>
- Thackway, W., Ng, M., Lee, C. L., & Pettit, C. (2023b). Implementing a deep-learning model using Google street view to combine social and physical indicators of gentrification. *Computers, Environment and Urban Systems*, 102, Article 101970. <https://doi.org/10.1016/j.compenvurbysys.2023.101970>
- Transparency International. (2024). Corruption perceptions index. available at <https://www.transparency.org/en/cpi/2023>. (Accessed 11 April 2024).
- Van Aken, J. E., & Romme, A. G. L. (2012). A design science approach to evidence-based management. In D. M. Rousseau (Ed.), *The oxford handbook of evidence-based management* (pp. 43–58). Oxford University Press. <https://doi.org/10.1093/oxfordhb/9780199763986.013.0003>
- Veuger, J. (2018). Trust in a viable real estate economy with disruption and Blockchain. *Facilities*, 36(1/2), 103–120. <https://doi.org/10.1108/F-11-2017-0106>
- Wang, Z., Wang, T., Hu, H., Gong, J., Ren, X., & Xiao, Q. (2020). Blockchain-based framework for improving supply chain traceability and information sharing in precast construction. *Automation in Construction*, 111, Article 103063. <https://doi.org/10.1016/j.autcon.2019.103063>
- Wendimu, D., & Biredagn, K. (2023). Developing a knowledge-based system for diagnosis and treatment recommendation of neonatal diseases. *Cogent Engineering*, 10(1). <https://doi.org/10.1080/23311916.2022.2153567>
- Werbach, K. (2018). *The blockchain and the new architecture of trust*. Cambridge, MA: MIT press. <https://doi.org/10.7551/mitpress/11449.001.0001>
- Wouda, H. P., & Opendakker, R. (2019). Blockchain technology in commercial real estate transactions. *Journal of Property Investment & Finance*, 37(6), 570–579. <https://doi.org/10.1108/JPIF-06-2019-0085>
- Wu, H., Zhong, B., Li, H., Chi, H., & Wang, Y. (2022). On-site safety inspection of tower cranes: A blockchain-enabled conceptual framework. *Safety Science*, 153, Article 105815. <https://doi.org/10.1016/j.ssci.2022.105815>
- Wyatt, P. (1996). The development of a property information system for valuation using a geographical information system (GIS). *Journal of Property Research*, 13(4), 317–336. <https://doi.org/10.1080/095999196368826>
- Wyatt, P. J. (1997). The development of a GIS-based property information system for real estate valuation. *International Journal of Geographical Information Science*, 11(5), 435–450. <https://doi.org/10.1080/136588197242248>
- Xu, Z., & Zou, C. (2021). What can blockchain do and cannot do? *China Economic Journal*, 14(1), 4–25. <https://doi.org/10.1080/17538963.2020.1748968>
- Yılmaz, O., Gürsoy Sürmeneli, H., & Alkan, M. (2023). Spatial-Land use planning system data model proposal for edition II of LADM. *Geocarto International*, 38(1), 1–21. <https://doi.org/10.1080/10106049.2023.2284278>
- Yang, R., Wakefield, R., Lyu, S., Jayasuriya, S., Han, F., Yi, X., Yang, X., Amarasighe, G., & Chen, S. (2020). Public and private blockchain in construction business process and information integration. *Automation in Construction*, 118, Article 103276. <https://doi.org/10.1016/j.autcon.2020.103276>
- Yapicioglu, B., & Leshinsky, R. (2020). Blockchain as a tool for land rights: Ownership of land in Cyprus. *Journal of Property, Planning and Environmental Law*, 12(2), 171–182. <https://doi.org/10.1108/JPEPL-02-2020-0010>
- Zhang, J., Lee, C. L., & Chan, N. (2019). The asymmetric effect of institutional quality on Chinese outward real estate investment. *International Journal of Strategic Property Management*, 23(6), 435–449. <https://doi.org/10.3846/ijspm.2019.10429>